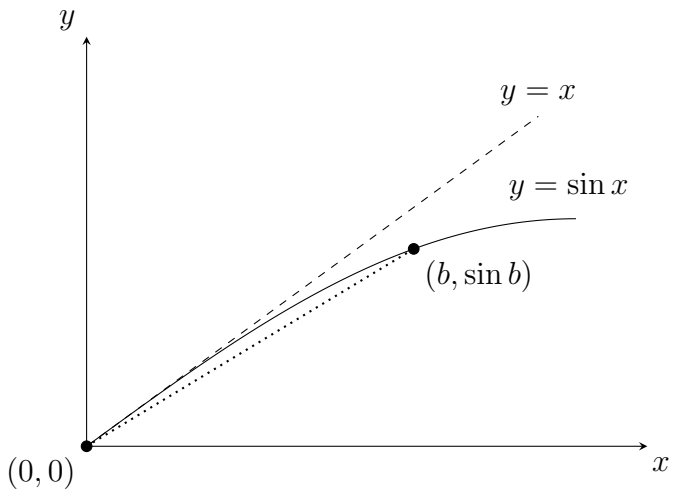
(i)



For $0 < x < b$,

$$\frac{x \sin b}{b} < \sin x < x$$

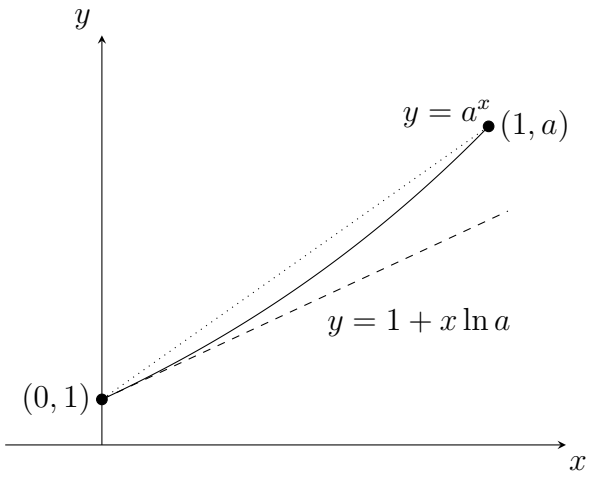
Area under chord $<$ Area under curve $<$ Area under tangent

$$\frac{1}{2} b \sin b < \int_0^b \sin x \, dx < \frac{1}{2} b^2$$

$$\frac{1}{2} b \sin b < 1 - \cos b < \frac{1}{2} b^2$$

$$1 - \frac{1}{2} b^2 < \cos b < 1 - \frac{1}{2} b \sin b$$

(ii)



Area under tagent $<$ Area under curve:

$$\int_0^1 a^x \, dx < \frac{a+1}{2}$$

$$\frac{a-1}{\ln a} < \frac{a+1}{2}$$

$$\frac{2(a-1)}{a+1} < \ln a$$

Area under tangent $<$ area under curve:

$$\int_0^1 a^x \, dx > \int_0^1 (1 + x \ln a) \, dx$$

$$\frac{a-1}{\ln a} > 1 + \frac{1}{2} \ln a$$

$$(\ln a)^2 + 2 \ln a < 2a - 2$$

$$(\ln a + 1)^2 < 2a - 1$$

$$\ln a < -1 + \sqrt{2a-1}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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